SECTION A

1. (a) A cantilever beam is subject to a uniformly distributed load q along CD as shown in Figure 1(a).

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

   (ii) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tensile** side. **(5 marks)**

   (b) A three-hinge frame is subject to one counter-clockwise point moment m applied on the right of pinned joint E, as illustrated in Figure 1(b).

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

   (ii) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tensile** side. **(5 marks)**

2. As shown in Figure 2(a), beam ABC is connected to truss member CD with a pinned joint C. And a uniformly distributed load q is applied along AB. The cross section of the beam is rectangular and the truss member is circular, see Figure 2(b) and Figure 2(c) for the dimensions, respectively.

   (a) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(3 marks)**

   (b) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tensile** side. **(3 marks)**

   (c) Find the maximum stress and the corresponding positions in the beam ABC. Express the stress in terms of q, L, E, b and h. **(3 marks)**

   (d) Find the deflection at point C in the beam ABC in terms of q, L, E and d. **(3 marks)**

   (e) Find the deflection at point B in the beam ABC in terms of q, L, E, b, h, and d. **(8 marks)**

3. Two frames, frame 1 and frame 2, are subject to point loads P_1 and P_2, as shown in Figure 3 (a) and (b) respectively.

   (a) Use Euler's buckling theory to calculate the critical buckling loads $P_{1,cr}$ and $P_{2,cr}$ of the two frames, respectively. For each of the structural members, EI = constant and $EA = \infty$. **(10 marks)**

   (b) How the critical buckling loads $P_{1,cr}$ and $P_{2,cr}$ will change when EA is reduced from infinity to zero? Explain the reasons. **(10 marks)**

Hint: According to Euler's formula, the critical load for a slender column with span L is $P_{cr} = \frac{\pi^2 EI}{(\mu L)^2}$, where μ is length coefficient. μ=1.0 for both pinned ends, μ=2.0 for one fixed end and one free end, μ=0.7 for one fixed end and one pinned end, and μ=0.5 for both fixed ends.

FIGURES

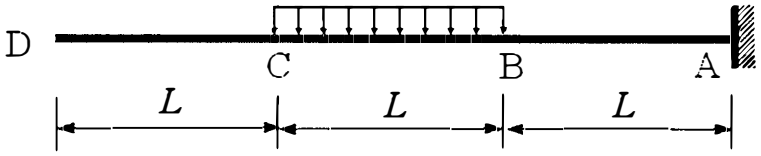


Figure 1(a) Beam

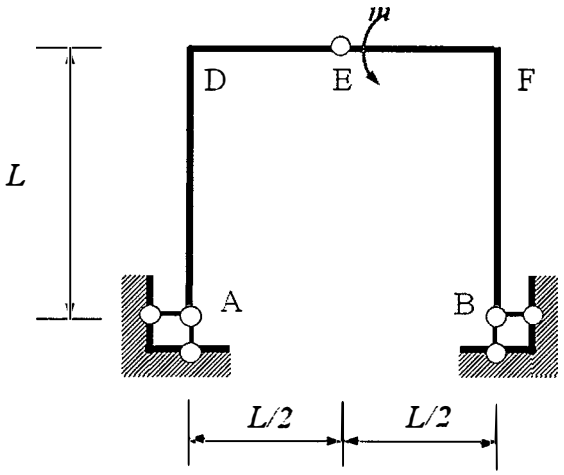


Figure 1(b) Three-Hinge Frame

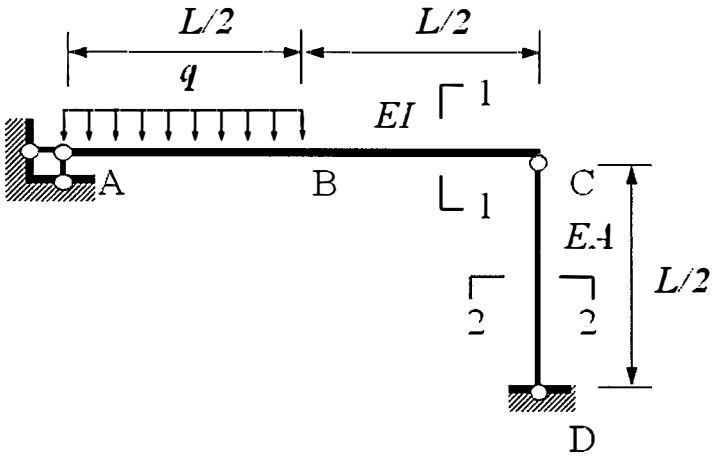


Figure 2(a) Frame ABCD

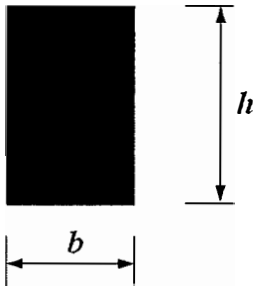


Figure 2(b) 1-1 Section

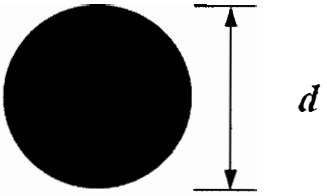


Figure 2(c) 2-2 Section

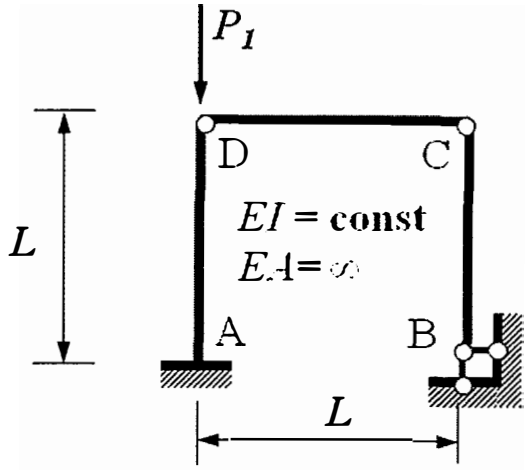


Figure 3(a) Frame 1

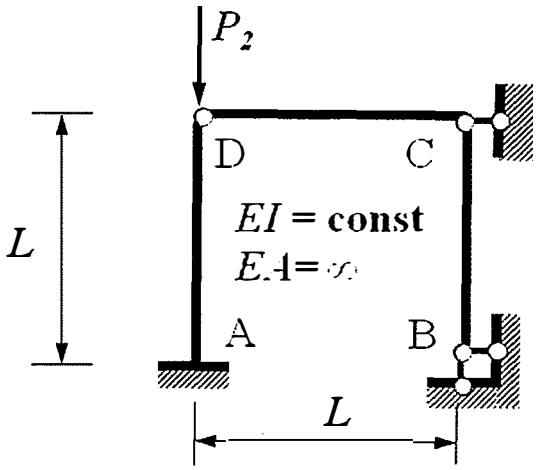


Figure 3(b) Frame 2

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